IVAN HERDIANTO

Bachelor of Informatics | Full-Stack Web Developer

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SUMMARY

Informatics fresh graduate from UPN "Veteran" Jawa Timur (GPA 3.89/4.00) specializing in full-stack web development using Laravel Framework. Built and delivered web-based applications with role-based access, SQL databases (MySQL, PostgreSQL), and map-based features (Leaflet.js), including a recruitment decision support system developed in Agile sprints during an internship. Also applied Python and machine learning (SVR, scikit-learn) in my thesis, deployed as a Streamlit web app.

EDUCATION

Universitas Pembangunan Nasional "Veteran" Jawa Timur 2022 – 2026

Faculty of Computer Science – Informatics | GPA: 3.89/4.00

- **Relevant Coursework:** Algorithms & Programming, Object-Oriented Programming, Web Programming, System Analysis & Design, Interface Design.
- **Undergraduate Thesis:** Estimation of Land Area Based On-Screen Digitization Utilizing Karney's Algorithm And Support Vector Regression

EXPERIENCE

PT. Mutiara Putri Gemilang – Internship January – April 2025

Full Stack Web Developer Intern

- **Built a Decision Support System (DSS) for employee recruitment** (Laravel, MySQL) using AHP and Profile Matching, replacing a manual, subjective selection stage with objective score-based ranking for >20 candidates per cycle.
- **Planned and ran Agile sprints** for 3+ core web application features, delivering all on time and within scope.
- **Coordinated** task allocation, deliverable tracking, and quality assurance with stakeholders to match their requirements.

Rakamin Academy – Independent Study Batch 7 (MSIB) September – December 2024

IT Full Stack Developer Trainee

- **Completed** a 4-month IT Full Stack Developer program (Web Development & Data Science) under Kampus Merdeka's MSIB.
- **Built a Waste Management Education Website** (Laravel, TailwindCSS, MySQL) with articles and interactive gamification, from UI/UX design in Figma through deployment. GitHub
- **Served as Front-End Developer**, building component-based interactive interfaces with responsive design and cross-browser compatibility.

PROJECTS

Land Parcel Area Estimation (Undergraduate Thesis) | Python, GeographicLib, scikit-learn, GeoPandas, Streamlit

- **Built an ML pipeline** on 400 Surabaya cadastral parcels: geodesic area via the Karney algorithm, morphological features (compactness, elongation, vertex count), and an SVR model tuned with GridSearchCV and 5-fold CV (MAPE 5.34%, R^2 0.9997).
- **Validated statistically** that geodesic (Karney) area is more accurate than planar area (MAPE 6.89% vs 8.41%, Wilcoxon $p = 0.0034$).
- **Deployed** as a Streamlit WebGIS app on Streamlit Cloud.

Delivery Tracking Unit System | Laravel, Alpine.js, TailwindCSS, MySQL, Leaflet.js GitHub

- **Built** a web system to manage and track vehicle unit deliveries: master data, orders, invoices, deliveries, and proof of delivery.
- **Implemented** role-based authentication (admin, driver, customer) with separate dashboards and real-time monitoring of delivery status, location, and coordinates.

Social Welfare Services Mapping Information System (SIMPELKESOS) | Laravel, Bootstrap, MySQL, Leaflet.js GitHub

- **Built** a web platform for resident data collection, socioeconomic mapping, and welfare analysis based on income deciles.
- **Implemented** a Leaflet.js map of Mojokerto Regency to visualize the distribution of socioeconomic conditions.

PUBLICATIONS

- **Author, JINTEKS 2026 (SINTA 4):** Manuscript on how polygon morphology (vertex count, compactness, elongation) affects the accuracy gap between planar and geodesic area calculation on 400 Surabaya cadastral parcels.
- **National proceeding paper:** "Perbandingan Estimasi Luas Lahan Hasil Digitasi Layar Menggunakan Algoritma Karney dan Random Forest."

SKILLS

Web & Backend	PHP, Laravel, JavaScript, Alpine.js, TailwindCSS, Bootstrap, HTML, CSS, Leaflet.js
Database	MySQL, PostgreSQL, SQL
Data & ML	Python, Pandas, scikit-learn (SVR, Random Forest), GeoPandas, GeographicLib, Streamlit
Tools	Git/GitHub, Figma
Also familiar with	Java, C/C++

CERTIFICATIONS & TRAINING

Junior Web Programmer – Certificate of Competency (BNSP) December 2025

- **Built a web-based e-commerce platform** with an admin backend offering full CRUD for inventory and data management.
- **Implemented role-based access control** (RBAC) to restrict access by user role, and an automated email invoicing system for customer transactions.

Google IT Support – Aguna Course (Coursera) January 2023

- IT support fundamentals: computer assembly, network configuration, software installation, and end-to-end troubleshooting using Linux, DNS, and CLI.

ORGANIZATIONS & COMMITTEES

HIMATIFA – Informatics Student Association March 2025 – January 2026

Head of Community Service Department

- **Led** technology-based community service programs, from planning and execution to evaluation, with the department team.
- **Managed partnerships** with external parties (schools, communities, institutions) and kept communication effective among stakeholders.

Abdi Desa 2024 July 2024

Vice Chair of the Organizing Committee

- **Initiated and coordinated** a technology-based community service program, tailoring materials and digital solutions to the real needs of village communities.
- **Improved digital literacy** through educational activities and hands-on practice, involving residents in using digital tools for productive activities and communication.